Nanjing University
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Birth Year: 1993

Junhao Liu

Research Interests

- Star formation, molecular clouds, magnetic fields, turbulence, filaments, jets and outflows

Employment

2026/1- **Nanjing University**, Nanjing, Jiangsu, China
Assistant Professor (tenure-track)

2023/9-2026/1 **National Astronomical Observatory of Japan (NAOJ)**, Mitaka, Tokyo, Japan
Project Researcher (postdoc), Host: Prof. Patricio Sanhueza and Prof. Fumitaka Nakamura

2021/6-2023/9 **East Asian Observatory (EAO)**, Hilo, Hawaii, USA
EAO Fellow (postdoc)

Research Experience

2018/8-2021/5 **Center for Astrophysics (CfA)** | **Harvard & Smithsonian**, Cambridge, MA, USA
SAO pre-doctoral Fellow, Supervisor: Dr. Qizhou Zhang

Education

2015/9-2021/3 **Nanjing University**, Nanjing, Jiangsu, China
Ph.D. in Astronomy. Supervisor: Prof. Keping Qiu

2011/9-2015/6 **Nanjing University**, Nanjing, Jiangsu, China
B.S. in Astronomy.

Mentoring Experience

2025- Xin Wu, PhD student at Xinjiang Astronomical Observatory, co-advised.
2024-2026 Yangjun Pu, undergraduate student at Nanjing University, co-advised.
2024-2026 Junjie Huang, undergraduate student at Nanjing University, co-advised.
2023-2025 Qiuyi Luo, PhD student at Shanghai Astronomical Observatory, co-advised.
2019-2020 Shixian Mo, undergraduate student at Nanjing University, co-advised.
2019-2020 Li Sun, PhD student at Purple Mountain Observatory, co-advised.

Approved PI Proposals (selected)

* **JCMT**

2023 "Magnetic Fields in Massive Star-Forming Regions". M24BP019.
2022 "Dust polarization survey of IRDCs". M22AD003.
2022 "Deviation from turbulence cascade in NGC 6334". M23AD001.
2021 "Are sub-virial clumps in IRDC G28.34 supported by magnetic fields?". M22AP018.
2018-2022 "Dust polarization survey of massive dense cores in Cygnus-X". M18BP047, M19BP037, M20AP048, M22AP019, and M22BP047.

* **SMA**

2019-2023 "Dust polarization survey of massive dense cores in Cygnus-X". 2019B-S028, 2021A-S008, 2022A-S040, and 2023A-S011.

* **ALMA**

2025 "Early fragmentation of massive protoclusters regulated by magnetic fields or turbulence? ". Cycle 12: 2025.1.00549.S.

2021 "Dust polarization survey of massive dense clumps in IRDCs". Cycle 8: 2021.1.01083.

Honors and Grants

2025 KAKENHI No.25K17445. 2,200,000 JPY. Grant-in-Aid for Scientific Research of JSPS

2022 Outstanding Young Scholar for the Molecular Clouds and Star Formation Colloquium

Academic Activities

2026/8 Conference Talk. Molecular Clouds and Star Formation Colloquium in Zhengding, China

2026/7 Conference Talk. Zhangheng Colloquium in Xiamen, China

2025/10 Conference Talk. International Symposium on Cosmic Magnetic Fields in Beijing, China

2025/08 Conference Talk. SFDE Conference in Quy Nhon, Vietnam

2025/08 Conference Talk. Molecular Clouds and Star Formation Colloquium in Emeishan, China

2025/04 Seminar Talk. NAOJ Science Seminar in Mitaka, Japan

2024/11 Invited Talk. Molecular Clouds and Star Formation Colloquium in Nanning, China.

2024/07 Conference Talk. East Asian ALMA Workshop in Seoul, Korea.

2024/03 Conference Talk. Bfields-2024 in Mitaka, Japan.

2023/08 Seminar Talk. NJU-SASS Science Seminar in Nanjing, China

2023/07 Seminar Talk. SHAO Science Seminar in Shanghai, China

2023/07 Seminar Talk. EAO Science Seminar in Hilo, USA

2023/02 Seminar Talk. Young Scholars Forum at Nanjing University (remote), Nanjing, China

2022/11 Seminar Talk. SWIFAR Seminar at Yunnan University (remote), Kunming, China

2022/09 Seminar Talk. KIAA-DoA Seminar at Peking University (remote), Beijing, China

2022/08 Conference Talk. Molecular Clouds and Star Formation Colloquium of China (remote).

2022/02 Seminar Talk. Shanghai Astronomical Observatory in Shanghai (remote), China

2021/07 Seminar Talk. EAO Science Seminar in Hilo (remote), USA

2021/06 Conference Talk. SOFIA/JCMT Magnetic Fields and the Structure of the Filamentary Interstellar Medium Online Workshop (remote)

2021/03 Flash Talk. IAU Symposium 360 ASTROPOL2020 in Hiroshima (remote), Japan

2019/11 Flash Talk. Harvard-Heidelberg Workshop on Star Formation in Cambridge, USA

2019/05 Poster Talk. Workshop on Polarization in Protoplanetary Disks and Jets in San Cugat, Spain

2018/08 Conference Talk. Molecular Clouds and Star Formation Colloquium in Lhasa, China

2017/11 Conference Talk. Molecular Clouds and Star Formation Colloquium in Yichang, China

2017/08 Conference Talk. Chinese Astronomical Society Meeting in Urumqi, China

Services

- Referee for ApJ, A&A, and PASJ.
- Reviewer of JCMT and ALMA proposals.

2024	LOC for Bfields-2024 conference in Mitaka, Japan.
2021-2023	EAO fellow's JCMT-related operation duties (50% of time)
2021-2023	Organizer of EAO science seminars.
2022	Co-organizer of the JCMT 2022 Users Meeting (remote). Chair for one session.

Observation Experience

2021/12	IRAM-30m remote observation. 1 night.
2018/02	Tianma-65m on-site observation. 5 days. Shanghai, China
2017/09	JCMT on-site observation. 7 nights. Mauna Kea, HI, US
2015/06	CSO remote observation. 5 days.

Skills

Software	CASA, STARLINK, IDL/MIR, MIRIAD, POLARIS, RADEX, GILDAS, RAMSES, Athena++
Data reduction	ALMA, SMA, NOEMA, JCMT, IRAM-30m, Tianma-65m, PMO-13.7m

Publications

ADS link to first-author/corresponding-author Papers

ADS link to all papers